MILITARY SPECIFICATION

SCREENS, HALFTONE, CONTACT, FOR PHOTOLITHOGRAPHY

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers photographic halftone contact screens used in the preparation of halftone negatives for photolithography.

1.2 Classification. The screens shall be magenta positive type and shall be of the following sizes and rulings, as specified (see 6.2):

Size 24- by 24-inch
Size 29- by 29-inch
Ruling 150-line
Ruling 300-line

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The contact screen shall consist of a uniform, photographically prepared grid of vignetted dots on a transport polyester film base. The screen shall be a magenta positive type.

3.2 First article . The contractor shall furnish one or more screens of each and ruling for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce screens that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Recycled material may be used (see 6.4) provided it meets the requirements of this specification.

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3.4 Size. The screens shall conform to the following dimensions:

	24 by 24 inch size	29 by 29 inch size
Working area, plus 0.5-, minus 0.0-inch	24- by 24-inch	29- by 29-inch
Overall size, plus 0.5-, minus 0.0-inch	25- by 25-inch	31- by 31-inch
Thickness, plus or minus 0.001-inch	0.009-inch	0.009-inch

3.5 Screen base. The film base, emulsion, and protective coating, if used, shall be free from any defects. Minute hairline scratches, specks or bubbles, flaws, emulsion flow lines, or discolorations that will not reproduce photographically (within the nominal overall size of the screened film area) when tested as specified in 4.5.2.1, or that are not larger or of greater frequency than specified in 3.6 shall be acceptable. If a border is provided outside the usable screened area of the film it shall be colorless, clear, and transparent, except for indentification marking.

3.6 Screen dot structure. The screen pattern shall consist of a grid of indentical vignetted dots uniformly spaced in rows inclined 45 degrees plus or minus 3 degrees, to the screen margins. Each dot shall evidence density graduations varying from maximum at the dot center and decreasing continuously to approximately transparent film at the interstice between each cluster of 4 dots. When exposures are made as specified in 4.5.2.1, the density gradient of the dot formation shall provide the required latitude and photographic response to reproduce all steps of a standard gray scale and shall permit extending, contracting, or shifting the reproduced gray scale dot size range. Irregularities shall not be larger than 0.010 inch in diameter, and shall not reproduce photographically when tested as specified in 4.5.2.1 and 4.5.2.2. In addition, irregularities shall comprise: (a) The complete absence of a dot: (b) the complete fill-in of the area between 4 dots, and (c) defects which are of sufficient frequency or magnitude to produce a visual variation to the unaided eye in a 1:1 photographic reproduction of the screen.

3.6.1 Screen ruling. The screen ruling shall be defined as the number of dots per lineal inch along the angle of dot formation. When the screen is measured as specified in 4.5.2.3, the actual dot count shall not vary by more than 5 percent from the commercial rating along any random selected row.

3.7 Flatness. When tested as specified in 4.5.2.1, the screen shall evidence no creases, indentations, or concave or convex areas that may induce wrinkle formation or out-of-contact areas.

3.8 Performance. When tested as specified in 4.5.2.2, the screen shall produce a uniform tint pattern over its entire usable screen area. The tint negative shall evidence no spots, lines, mottled areas, or tonal differences

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caused by and traceable to the screen other than the defects permitted in 3.6, and the corresponding defects inherent in the exposure and processing of the film. When tested as specified in 4.5.2.1, the screen shall also reproduce or modify the tonal gradation of a density step wedge or equivalent copy.

3.9 Instructions. Instructions for the proper care, handling, and cleaning of the screen shall be printed on the outside of the inner container. Instructions for the use of the screen shall be included in each unit package. The screen shall be indentified by permanently printing the following information in the margin of the emulsion side of the screen:

Contact Screen
Screen Side

3.10 Filters. One filter of commercial designation No. 4 yellow, and one No. 30 rose shall be provided with each contact screen. Filters shall be not less than 2 inches square.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Governement reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First-article screen inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-article inspection.

4.3.1 Examination. The first-article screen shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-article screen shall be tested as specified in 4.5.2.1 throught 4.5.2.3. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Each screen shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual. Each screen shall be tested as specified in 4.5.2.3. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The screens shall be examined as specified herein for the following defects:

101. Improper size or thickness.
102. Hairline scratches, specks or bubbles, flaws, emulsion flow lines, or discolorations not as specified.
103. Screen dot structure not as specified.
104. Screen ruling not as specified.
105. Creases, indentations, or wrinkles not as specified.
106. Spots, lines, or mottled areas not as specified.
107. Missing or improper indentification markings on screen or containers.
108. Instructions not included on inner containers.
109. Filters not provided.

4.5.2 Tests.

4.5.2.1 Camera test. A process copying camera equipped with a anastigmatic lens, vacuum back, and a copyboard illuminated with pulsed xenon lights shall be used. The vacuum back shall be plane and true over contact screen area to within 0.002 inch. Tests to determine conformance to 3.7 shall be made with a dummy film of maximum screen capacity that is first applied to the camera vacuum back and over which the screen is placed. Where possible, the vacuum back shall be of the suction type operating through a multiple grid of small openings or a porous plate structure with a suction differential of approximately one pound per square inch. If the suction type is not available, the high vacuum grooved type may be used. Tests to determine conformance to 3.6 shall be made with a commercial paper gray scale having not less than 10 steps ranging in density from approximately 0.1 to 2.0, or with other copy having a similar range of graduated tone values. Successive exposures shall be made under the conditions specified on different films or

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on the same film by an appropriate shift of masking on the camera back with corresponding image displacement. Film shall be processed in accordance with the recommendations of the manufacturer of the film. Exposures shall be made to provide: (a) Normal contrast approximately reproducing the tone values of all the stops ranging from a fine pinpoint transparent dot in the white or highlights to a similar pinpoint dot for the black or deep shadow areas of the copy; (b) Extended contrast whereby the central portion of the gray scale range of tones is reproduced to provide approximately the same range of clear and opaque pinpoint dots obtained at the extreme for normal contrast: and (c) Condensed contrast in which the reproduction of the gray scale or copy is reduced to provide approximately a 15 to 25 percent dot corresponding to the deep shadow (black) area ranging to a similar 15 to 25 percent transparent dot corresponding to the highlight (white) area. Finally, by an adjustment of exposure as directed by the instructions, the condensed range shall be shifted to both ends of the scale successively to reproduce in a range from pinpoint opening or opaque dot to approximately 50 percent dot area. All dots so formed shall be sharply defined with continuous edges (except along boundaries of density changes) when they are viewed under not less than 10-diameter magnification, and the transparent areas shall be clear with necessary allowances for the normal film characteristics as regards grain, contrast, and film base fog level. Nonconformance to 3.6, 3.7, or 3.8 shall constitute failure of this test.

4.5.2.2 Contact exposure. Exposures for testing overall uniformity of screen texture reproduction can be accomplished in the camera by use of a flashlamp over the lens and extending the camera bellows until the lens-to-screen distance is not less than twice the diagonal of the nominal screen size; or its equivalent obtained by using a contact vacuum printing frame with a concentrated diffused light source such as an opal lamp bulb in a nonreflecting holder at a point incident to the perpendicular line from the screen center and at the same light source-to-screen distance. Test exposure shall first be made to obtain approximately a 25-percent tint. A film covering the usable screen area shall then be exposed for such a tint and processed in accordance with the manuufacturer's instructions. The resulting film negative shall be examined under the same conditions as those for the screen in 4.5.2.1 to determine conformance to 3.8. All visible defects and variances of dot size shall be circled on the film or on an attached overlay sheet with a china marking pencil. Marked defects shall be referenced to the contact screen to determine if they were produced by the screen and if so, whether the defects exceed the requirements specified in 3.6. Nonconformance to 3.6 or 3.8 shall constitute failure of this test.

4.5.2.3 Screen ruling and angle. The screen shall be tested with an optical projector or microscrope by direct count against a calibrated scale on the projected screen or to a calibrated reticle in the microscope to ascertain the number of dots per lineal inch along the angle of dot formation. In lieu of the direct count, a check screen or grating on glass may be used to directly

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indicate agreement or extent of variance by the interference pattern formed throught contact examination by transmitted light. The screen angle shall be measured with a plastic protractor accurate to plus or minus 0.5 degree. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.2 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

110. Materials, methods or containers not as specified for Level A or B. Each incorrect material, method or container shall constitute one defect.
111. Screens of unlike size or description placed together for Level A or Commercial.
112. Screens not preserved in accordance with the referenced document for Commercial.
113. Screens of unlike size or description packed together for Level A, B, or Commercial.
114. Quantities of packed screens not as specified for Level A or B.
115. Marking missing, illegible, incorrect, or incomplete for for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each screen in the vendor's package shall be preserved in accordance with MIL-P-116, Method IC-1. When a vendor's package is not furnished, each screen shall be placed between fiberboard stiffeners and preserved in accordance with MIL-P-116, Method IC-1.

5.1.1.1 Consolidation. Screens of like size and description shall be consolidated together in quantities of 10 in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be closed in accordance with the appendix to the box specification, Method V.

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5.1.2 Commercial. Screens shall be preserved in accordance with MIL-STD-1188. Only screens of like size and description shall be placed together.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Screens of like size and description, preserved as specified in 5.1, shall be packed in quantities of 10 consolidated containers in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Screens shall be packed as specified in 5.2.1 for Level A except that boxes shall conform to PPP-B-636, W5c, style optional.

5.2.3 Commercial. Screens of like size and description, preserved as specified in 5.1, shall be packed together in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (Level A or Level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The contact printing screen is intended to be used in the preparation of halftone negatives.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and ruling required (see 1.2).
(c) Time frame required for submission of first-produced screen and number of screen required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First-produced screen. Any changes or deviations of production screens from the approved first-produced screen during production will be subject to the approval of the contracting officer. Approval of the first-produced screen will not relieve the contractor of his obligation to furnish screens conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used when practical, provided it meets the requirements of this specifications (see 3.3)

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6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor maintenance and operation manual to be furnished with each screen.

Custodian:
Army - ME

Review activity
DLA - GS

Preparing activity:
Army - ME

Project 3610-A309

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-S-40034B(ME) Screens, Halftone, Contact, for Photolithography

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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